सहायक प्राध्यापक (सूक्ष्मजीवशास्त्र) महाराष्ट्र शिक्षण सेवा, गट-अ
(महाविद्यालयीन शाखा)
उच्च व तंत्र शिक्षण विभाग
Assistant Professor (Microbiology) Maharashtra Education Service, Group-A
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परीक्षेचे टप्पे: १) लेखी परीक्षा - २०० गुण २) मुलाखत- ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयांशी संबंधीत घटक (११४५)	इंग्रजी	१००	२००	एक तास	पदव्युत्तर पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयांशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

अ.क्र.	घटक व उपघटक
1	**General Microbiology** • History and scope of microbiology • Principles and methods of taxonomy, classification and taxonomy of microorganisms • Microscopy and Staining Techniques: Types of microscopy (light, electron, fluorescence), Basic staining techniques: simple, differential (Gram, acid-fast) • Control of Microbial Growth: Physical methods: heat, filtration, radiation; Chemical control: disinfectants, antiseptics, and antibiotics. • Structure and function of microbial cells (bacteria, archaea, fungi, viruses, algae, and protozoa) • Growth and reproduction of microbes, including microbial growth kinetics
2	**Microbial Biochemistry, Physiology and Metabolism** • Principles of biophysical chemistry (pH, buffer, reaction kinetics, thermodynamics and colligative properties) • Structure and function of biomolecules in microbes • Conformation and stability of proteins and nucleic acids. • Overview of microbial metabolism: catabolic and anabolic pathways • Metabolic pathways and energy production in microbial cells, electron transport chain and ATP synthesis • Lipid, nucleic acid, amino acid and protein metabolism in microbes • Aerobic and anaerobic respiration, fermentation, and phototrophy • Structure and function of enzymes, mechanism of enzyme action and enzyme kinetics, regulation of enzyme activity • Biochemical adaptations of extremophiles • Secondary metabolites and bioactive compounds

	• Structural biology tools (e.g., X-ray crystallography, NMR, cryo-EM) in studying microbial proteins • Bioanalytical methods: Spectroscopic, Chromatographic and Electrophoretic
3	**Evolution, Microbial Genetics and Molecular Biology** • Mendelian principles, concept of gene and extension of Mendelian principles, Lamarck, Darwin- concepts of variation, adaptation, struggle, fitness and natural selection. • Origin of cells and unicellular evolution: origin of basic biological molecules, abiotic synthesis of organic monomers and polymers, concept of Oarin & Hldane, experiment of Miller, evolution of prokaryotes, origin of eukaryotic cells. • Salient features of genetic code, structure, replication, and function of DNA and RNA • Molecular evolution: molecular divergence and molecular clocks. • Gene mapping methods • Gene expression, operon concept and regulation in prokaryotes and eukaryotes • RNA synthesis and processing • Protein synthesis and processing • Genetic recombination and implications for microbial evolution; horizontal gene transfer: transformation, transduction, and conjugation; Plasmids, transposons • Extra chromosomal inheritance • Mutations and repair mechanisms • Overview of cancer genetics, immunogenetics and pharmacogenomics • Basic and advanced genetic engineering techniques, Microbial genomics and proteomics
4	**Immunology and Host-Microbe Interactions** • Blood and circulation • Components and functions of the immune system: innate and adaptive immunity • Antigens: Determinants of antigenicity, types of antigens, MHC • Antibodies: structure, types, classes with functions. • Antigen-antibody reactions and immune responses to infections • Hypersensitivity, autoimmunity and autoimmune diseases. • Mechanisms of microbial pathogenicity: an overview of virulence factors, persistent infections, bacterial toxins, enzymes, and surface structures in pathogenesis. • Host-microbe interactions: commensalism, mutualism, parasitism • Mechanisms of microbial evasion of host defences and immune evasion • Histochemical and immune techniques: detection of molecules using ELISA, RIA, Western Blot, immunoprecipitation, flowcytometry and immunofluorescence microscopy, detection of molecules in living cells, in situ hybridization (FISH & GISH)
5	**Medical Microbiology and Infectious Diseases** • Microbial pathogens: bacterial, viral, fungal, and parasitic diseases • Emerging infectious diseases and zoonoses • Antimicrobial agents, mechanisms of antibiotic actions and resistance, and current strategies in antibiotic discovery and development • Vaccines, Toxoids, immune sera, immunization schedule • Epidemiology, prevention, and control of infectious diseases
6	**Environmental and Soil Microbiology** • The Environment: Physical environment, biotic environment, biotic and abiotic interactions • Microbial ecology: interactions and roles in ecosystems, Biofilms • Adaptive evolution and genome plasticity in microbes • Biogeochemical cycles: carbon, nitrogen, sulphur, and phosphorus • Microbial interactions with plants: the rhizosphere and plant growth promotion • Microbial communities in soil, water, and extreme environments • Bioremediation and microbial applications in environmental clean-up
7	**Industrial, Pharmaceutical and Food Microbiology** • Microbial fermentation processes and applications • Production of bio-based products: enzymes, antibiotics, biofuels, organic acids • Microbial spoilage of food and food preservation techniques, Pasteurisation and other heat treatments, foodborne pathogens and foodborne illnesses • Methods for detecting and quantifying microorganisms in food and milk • Principles of Hazard Analysis and Critical Control Points (HACCP) in the food and dairy

	industry, Overview of GMP for the food industry • Probiotics, prebiotics, functional foods and their role in health • Overview of the drug life cycle: Stages of drug development, Clinical Research • Quality control, assurance and safety regulations in the food and pharmaceutical industries
8	**Virology and Viral Pathogenesis** • Structure and classification of viruses • Replication strategies of DNA and RNA viruses • Viral pathogenesis and mechanisms of disease production • Antiviral strategies and vaccine development • Emerging viral pathogens and zoonotic viruses
9	**Biotechnology and Genetic Engineering** • Techniques in recombinant DNA technology: cloning, gene editing, and expression • Applications of biotechnology in medicine, agriculture, and industry • Microbial Bio-mining, biofertilizers, biopesticides • Microbial biopolymers and bioplastics • Principles of synthetic biology and metabolic engineering
10	**Bioinformatics and Data Analysis in Microbiology** • Introduction to bioinformatics databases and sequence alignment • Phylogenetic analysis and microbial classification • Omics technologies: genomics, transcriptomics, proteomics, and metabolomics • Analysis of microbial communities using metagenomics Basic statistical and computational tools for microbiological data analysis
11	**Emerging Topics and Trends in Microbiology** • Current advances in microbiome research and human health • Molecular approaches to diagnosis and strain identification • CRISPR and gene editing technologies in microbiology • Bioterrorism and high-risk pathogens • Bioethics of microbial research, biosafety levels, and regulations • Systems biology and applications in understanding microbial networks • Environmental microbiology and implications for climate change

दिनांक – ३० एप्रिल, २०२५

अवर सचिव
महाराष्ट्र लोकसेवा आयोग